

COGNIZANT DIGITAL NURTURE

Java Full Stack Program (Java FSE)

WEEK 1 ASSIGNMENTS: CODE & OUTPUT

Submitted by:

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1. Design Patterns & Principles (Singleton Pattern)

Source Code:

```
class Logger {
    private static Logger instance;
    private Logger() { System.out.println("Logger instance created."); }
    public static synchronized Logger getInstance() {
        if (instance == null) { instance = new Logger(); }
        return instance;
    }
    public void log(String message) { System.out.println("[LOG] " + message); }
}

public class SingletonPatternExample {
    public static void main(String[] args) {
        Logger logger1 = Logger.getInstance();
        logger1.log("First log message.");
        Logger logger2 = Logger.getInstance();
        logger2.log("Second log message.");
        System.out.println("Are both instances the same? " + (logger1 == logger2));
    }
}
```

Execution Output:

```
=== Singleton Pattern ===
Logger instance created.
[LOG] First log message.
[LOG] Second log message.
Are both instances the same? true
```

2. Data Structures & Algorithms (Inventory Management)

Source Code:

```
import java.util.HashMap;

class Product {
    private int productId;
    private String productName;
    public Product(int productId, String productName) {
        this.productId = productId;
        this.productName = productName;
    }
    public int getProductId() { return productId; }
    public String getProductName() { return productName; }
}

class Inventory {
    private HashMap<Integer, Product> products = new HashMap<>();
    public void addProduct(Product product) {
        products.put(product.getProductId(), product);
        System.out.println("Added: " + product.getProductName());
    }
}

public class InventoryManagement {
    public static void main(String[] args) {
        Inventory inv = new Inventory();
        inv.addProduct(new Product(101, "Laptop"));
    }
}
```

Execution Output:

```
=== Inventory Management ===
Added: Laptop
```

3. PL/SQL Database Programming (Customer Package)

Source Code:

```
CREATE TABLE Customers (  
    CustomerID NUMBER PRIMARY KEY,  
    Name VARCHAR2(100),  
    Balance NUMBER  
);  
  
CREATE OR REPLACE PACKAGE CustomerManagement AS  
    PROCEDURE AddCustomer(p_id IN NUMBER, p_name IN VARCHAR2, p_bal IN NUMBER);  
END CustomerManagement;  
/  
  
CREATE OR REPLACE PACKAGE BODY CustomerManagement AS  
    PROCEDURE AddCustomer(p_id IN NUMBER, p_name IN VARCHAR2, p_bal IN NUMBER) IS  
    BEGIN  
        INSERT INTO Customers (CustomerID, Name, Balance) VALUES (p_id, p_name, p_bal  
    );  
        COMMIT;  
    END AddCustomer;  
END CustomerManagement;  
/
```

Execution Output:

Procedure CUSTOMERMANAGEMENT compiled.
Package Body CUSTOMERMANAGEMENT compiled.

4. Test Driven Development (JUnit Calculator Tests)

Source Code:

```
package com.cognizant.basic;

import static org.junit.jupiter.api.Assertions.*;
import org.junit.jupiter.api.BeforeEach;
import org.junit.jupiter.api.Test;

public class CalculatorTest {
    private Calculator calculator;

    @BeforeEach
    public void setUp() {
        calculator = new Calculator();
    }

    @Test
    public void testAdd() {
        assertEquals(30, calculator.add(10, 20));
    }
}
```

Execution Output:

```
[INFO] Running com.cognizant.basic.CalculatorTest
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] Results:
[INFO] Tests run: 68, Failures: 0, Errors: 0, Skipped: 0
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
```

5. Logging Framework (SLF4J & Logback Configuration)

Source Code:

```
<configuration>
  <appender name="console" class="ch.qos.logback.core.ConsoleAppender">
    <encoder>
      <pattern>%d{HH:mm:ss.SSS} [%thread] %-5level %logger{36} - %msg%n</pattern>
    </encoder>
  </appender>
  <root level="debug">
    <appender-ref ref="console" />
  </root>
</configuration>
```

Execution Output:

```
11:36:55.920 [main] DEBUG c.c.logging.AppendersLoggingExample - Debug log initialized
.
11:36:55.921 [main] INFO c.c.logging.AppendersLoggingExample - System process started.
```